

OIL-Immersed TR PM FORM

EQ#: _____ Unit: _____ SN# _____ WO# _____ Date: _____ PM Cycle: 2190 Days or each T/A

EQ	PM Item	Standard	Result	Reading/ Comment												
TR Main Body	1 Winding Insulation Resistance Measurement	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 33%;">Rated Voltage (Volts)</th> <th style="width: 33%;">Minimum DC Test Voltage (VDC)</th> <th style="width: 33%;">Recommended Minimum Insulation Resistance (MΩ)</th> </tr> </thead> <tbody> <tr> <td>0 - 600 V</td> <td>1000 VDC</td> <td>100 MΩ</td> </tr> <tr> <td>601 - 5000 V</td> <td>2500 VDC</td> <td>1000 MΩ</td> </tr> <tr> <td>>5000 V</td> <td>5000 VDC</td> <td>5000 MΩ</td> </tr> </tbody> </table>	Rated Voltage (Volts)	Minimum DC Test Voltage (VDC)	Recommended Minimum Insulation Resistance (MΩ)	0 - 600 V	1000 VDC	100 MΩ	601 - 5000 V	2500 VDC	1000 MΩ	>5000 V	5000 VDC	5000 MΩ	PASS ☐ Abnormal ☐	
Rated Voltage (Volts)	Minimum DC Test Voltage (VDC)	Recommended Minimum Insulation Resistance (MΩ)														
0 - 600 V	1000 VDC	100 MΩ														
601 - 5000 V	2500 VDC	1000 MΩ														
>5000 V	5000 VDC	5000 MΩ														
2 Dielectric Frequency Response (DFR) Test	≥230kV: ≤1.25% 69–230kV: ≤2.0% ≤69kV: ≤3.0%	PASS ☐ Abnormal ☐														
3 Measurement of Dielectric Power Factor (tanδ)	≤1.0% (at 20°C or 68°F)	PASS ☐ Abnormal ☐														
4 Winding Resistance Measurement	Deviation within ±5% among three-phase resistance and average value	PASS ☐ Abnormal ☐														
5 Excitation Current Test	Within ±30% compared to commissioning test value (if exceeded, check test voltage and residual magnetism)	PASS ☐ Abnormal ☐														
6 Bushing Cleaning & Inspection	Surface must be clean, free of dust and salt; no abnormal hot spot marks on contact point	PASS ☐ Abnormal ☐														
7 Busbar Connections	No oxidation, bolts tightened, bare parts free of verdigris	PASS ☐ Abnormal ☐														
8 Exterior Cleaning & Inspection	No oil leakage, rust, or paint peeling (touch up if needed)	PASS ☐ Abnormal ☐														
9 Cooling System	No rust or oil leakage, cooling fans operate normally, no abnormal noise or vibration	PASS ☐ Abnormal ☐ N/A ☐														
10 Nitrogen Pressure	Follow “Nitrogen Pressure–Oil Temperature” curve	PASS ☐ Abnormal ☐ N/A ☐														
11 Breather	Replace desiccant, oil level in oil cup must be higher than breathing port (renew oil if needed)	PASS ☐ Abnormal ☐ N/A ☐														
12 Conservator Tank	Oil level follows “Oil Level–Oil Temperature” curve, no leakage or rust	PASS ☐ Abnormal ☐ N/A ☐														
13 Grounding Conductor	No breakage, no loosening, clamping point free of corrosion and good contact	PASS ☐ Abnormal ☐														
Neutral	1 Internal & External Cleaning	No dust, no rust, bolts tightened, wiring and components free of overheating/discoloration	PASS ☐ Abnormal ☐ N/A ☐													

Grounding Resistor Box (NGR)	2	Neutral Grounding Resistance Measurement	Measured resistance value must meet OEM standard	PASS ☐ Abnormal ☐ N/A ☐	
	3	Label Inspection	Grounding resistor data must be clearly marked on panel	PASS ☐ Abnormal ☐ N/A ☐	
Grounding Resistance	1	Ground Resistance Measurement	Ground resistance must be < 5 Ω	PASS ☐ Abnormal ☐	
On-Load Tap Changer (OLTC)	1	Internal and External Inspection (Outsourced)	Per manufacture's standard	PASS ☐ Abnormal ☐ N/A ☐	
Terminal Box	1	Internal Cleaning	No water accumulation, no dust, box sealing intact	PASS ☐ Abnormal ☐	
Local Control Panel	1	External Inspection	Indicator lights working, labels intact, no dust, no rust	PASS ☐ Abnormal ☐ N/A ☐	
	2	Internal Inspection	No dust, no rust, bolts tightened	PASS ☐ Abnormal ☐ N/A ☐	
			Cooling fans operate normally, no abnormal noise	PASS ☐ Abnormal ☐ N/A ☐	
			Terminal connections free of oxidation/loosening, no overheating or discoloration	PASS ☐ Abnormal ☐ N/A ☐	
			Control circuits function normally	PASS ☐ Abnormal ☐ N/A ☐	
			After check, return NFB to ON position	PASS ☐ Abnormal ☐ N/A ☐	
Protective Relays	1	Buchholz Relay Inspection	Air release and test cocks tightened; no gas present in oil level window, if gas detected, release and investigate root cause	PASS ☐ Abnormal ☐ N/A ☐	
	2	Sudden Pressure Relay Inspection	No visible damage, test rod intact	PASS ☐ Abnormal ☐ N/A ☐	
	3	Pressure Relief Device Inspection	Protective cover not opened, center pin not ejected/broken	PASS ☐ Abnormal ☐ N/A ☐	
	4	Control Circuit	Test tripping and alarm interlocks for Buchholz, sudden pressure relay, and relief device	PASS ☐ Abnormal ☐ N/A ☐	
Thermometer	1	Appearance and Function Check	No damage, no fogging, contacts free of corrosion	PASS ☐ Abnormal ☐	
			Cooling fan starts at 50°C or 122 °F (stage 1), 75°C or 167 °F (stage 2). Over-temperature alarm output at 90°C or 194 °F Trip contact output at 120 °C or 248 °F.	PASS ☐ Abnormal ☐	
	2	Control Circuit	Test tripping and alarm interlock functions	PASS ☐ Abnormal ☐	
Cable Box	1	Box Inspection	Cable clamping device tightened and secure	PASS ☐ Abnormal ☐	
			Cable sealing compound intact and watertight	PASS ☐ Abnormal ☐	
			Soft copper tape not straightened	PASS ☐ Abnormal ☐	
			No oil leakage at insulators	PASS ☐ Abnormal ☐	

			Temp-indicator label not discolored	PASS ☐ Abnormal ☐	
			Box free of rust/water ingress	PASS ☐ Abnormal ☐	
Bushings	1	Bushings Inspection	No cracks, no damage, no black contamination	PASS ☐ Abnormal ☐	
			Oil level above center line of sight glass	PASS ☐ Abnormal ☐	
			Terminals tightened	PASS ☐ Abnormal ☐	
	2	Bushings Measurement (≥69kV)	Measure insulation resistance	PASS ☐ Abnormal ☐ N/A ☐	
			Measure power factor (tanδ)	PASS ☐ Abnormal ☐ N/A ☐	
			Measure capacitance	PASS ☐ Abnormal ☐ N/A ☐	
Busbars	1	Busbar Inspection	No discoloration, deformation, or verdigris corrosion at joints	PASS ☐ Abnormal ☐	
			Bolts tightened, no loosening or abnormal blackening	PASS ☐ Abnormal ☐	

Action:

Rev. 0

Manager/ Engineer Review By:_____ Supervisor Review:_____ Inspect By:_____